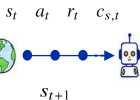


1 Rollout

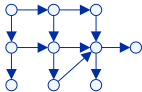


observed
episode cost
 $\bar{J}_k$

Collect transitions
($s_t, a_t, r_t, c_{k,t}, s_{t+1}$)

2 Flow inputs

Rollout graph



flow $\hat{F}$



adv. A



cost C_k

3 Constraints

SOFT

$$\langle C_k, F \rangle \leq b_k$$
$$b_k = B_k / L$$

HARD

$F_H = 0$
(excluded edges)

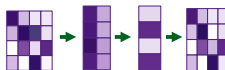
limits for $F^\star$

4 Flow projection

$$\hat{F} \rightarrow F^\star$$

entropic projection

$$\hat{F} \rightarrow F^\star$$



target flow; policy executes π

safety prices $\bar{\lambda}_k^\star$

5 Priced actor

prices + feedback

$$\lambda_k = \bar{\lambda}_k^\star + \beta_k$$

$$\beta_k \leftarrow [\beta_k + \alpha e_k]_+$$

$$e_k = \bar{J}_k - B_k$$

$$\tilde{A} = (A_r - \sum_k \lambda_k A C_k) / (1 + \sum_k \lambda_k)$$

PPO / TRPO $\rightarrow \pi_{t+1}$



next policy